29. ROLLING SHUTTER :

General :

Rolling shutters shall conform to IS : 6248. These shall include necessary locking arrangement and handles etc. These shall be suitable for fixing in the position as specified ie., outside or inside on or below lintel or between jambs of the opening. The door shall be either push and pull type or operated with mechanical device supplied by the firm. Shutters upto 10 sq metre shall be of push and pull type and shutters with an area of over 10 sq metre shall generally be provided with reduction gear operated by mechanical device with chain or handle, if bearings are specified for each of operation, these shall be paid for separately.

Shutter:

The shutter shall be built up of inter-locking lath section formed from cold rolled steel strips. The thickness of the sheets from which the lath sections have been rolled shall be not less than 0.90 mm for the shutters upto 3.5 m width and not less than 1.20 mm for shutters above 3.5 m width. Shutters above 9 metres in width should be divided in 2 parts with provision of one middle fixed or movable guide channel or supported from the back side to resist wind pressure. The lath section shall be rolled so as to have interlocking curls at both edges and a deep corrugation at the centre with a bridge depth of not less than 12 mm to provide sufficient curtain of stiffness for resisting manual pressures and normal wind pressure. Each lath section shall be continuous single piece without any welded joint. When interlocked, the lath sections shall have a distance of 75 mm rolling centres. Each alternate lath section be fitted with malleable cast iron or mild steel clips securely rivetted at either ends, thus locking the lath section at both ends and preventing lateral movement of the individual lath sections. The clips shall be so designed as to fit the contour of the lath sections.

Spring:

The spring shall be of coiled type. The spring shall be manufactured from high tensile spring steel wire or strips of adequate strength conforming to 18 : 4454-Part I.

Roller and Brackets: The suspension shaft of the roller shall be made of steel pipe conforming to heavy duty as per 18:1161. For shutter upto 6 metre width and height not exceeding 5 metre, steel pipes of 50 mm nominal bore shall be used. The shaft shall be supported on mild steel brackets of size 375 x 375 x 3.15 mm for shutters upto a clear height of 3.5 metre. The size of mild steel brackets shall be 500 x 500 x 10 mm for shutters of clear height above 3.5 m and upto 6.5 m. The suspension shaft clamped to the brackets shall be fitted with rotatable cast iron pulleys to which the shutter is attached. The pulleys and pipe shaft shall be connected by means of pretensioned helical springs to counter balance the weight of the shutter and to keep the shutter in equilibrium in any partly open position.

When the width of the opening is greater than 3.5 mtr. the cast iron pulleys shall be interconnected with a cage formed out of mild steel flats of at least 32 x 6 mm and mild steel dummy rings made of similar flats to distribute the torque un~formly. 8elf aligning two row bMi bearing with special cast iron casings shall be provided at the extreme pulley and caging rings shall have a minimum spacing of 15 mm and at least 4 number flats running throughout length of roller shall be provided.

In case of shutters of large opening with mechanical device for opening the shutter the roller shall be fitted with a purion wheel at one end which in contact with a worm fitted to the bracket plate, caging

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and pully with two ball bearing shall be provided.

Guide Channel:

The width of guide channel shall be 25 mm the minimum depth of guide channels shall be as follows:

Clear width of shutters	Depth of guide channel
Upto 3.5 m	65mm
3.5 m upto 8 m	75mm
8 m and above	100mm

The gap between the two legs of the guide channels shall be sufficient to allow the free movement of the shutter and at the same time close enough to prevent rattling of the shutter due to wind.

Each guide channel shall be provided with a minimum of three fixing cleats or supports for attachment to the walls or column by means of bolts or screws. The. spacing of cleats shall not exceed 0.75 m. Alternatively, the guide channels may also be provided with suitable dowels, hooks or pins for embedding in the walls.

The guide channels shall be attached to the jambs, plumb and true either in the overlapping fashion or embedded in grooves, depending on the method of fixing.

Cover : Top cover shall be of mild steel sheets not less than 0.90 mm thick and stiffened with angle or flat stiffeners at top and bottom edges to retain shape

Lock plates with sliding bolts, handles and anchorfng rods shall be as per IS: 6248.

Fixing

The arrangement for fixing in different situations in the opening shall be as per IS : 6248.

Brackets shall be fixed on the lintel or under the lintel as specified with rawl, plugs, and screws bolts etc. The shaft along with the spring shall then be fixed on the brackets.

The lath portion (shutter) shall be laid on ground and the side guide channels shall be bound with ropes etc. The shutter shall then be placed in position and top fixed with pipe shaft with bolts and nuts. The side guide channels and cover frames shall then be fixed to the walls through the plate welded to the guides. These plates and bracket shall be fixed by means of steel screws bolts, and rawl plugs concealed in plaster to make their location invisible. Fixing shall be done accurately in a workmen like manner that the operation of the shutter is easy and smooth.

Rolling Grills

Rolling grill is meant to provide visibility or ventilation or both, the degree of protection and safety is less as compared to a rolling shutter. The situations where a certain amount of ventilation combined with safety is required rolling shutter-cum-grill may be provided in which the rolling shutter may have rolling grill portion either at the top or at the bottom or at both places. In addition, the rolling grill portion may also be provided in the middle of the shutter. The total height of the grill portion in all the segments of rolling shuttercum-grill shall not exceed 0.1 m and the height of the grill portion in any

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individual segment shall not be more than 0.5 m.

Rolling grills are similar in design, construction and operation to rolling shutters and all the provisions of rolling shutter shall be applicable to rolling grills except in respect of the shutter portion, and shall conform to IS : 6248.

Shutters

Rolling grill shutter and the rolling grill portion of the rolling shutter-cum-grill shall be fabricated with 8 mm diameter mild steel round bars. Straight bars and bars bent to the required profile are placed alternatively and held in position with 20 mm wide and 5 mm thick mild steel flat links. Straight bars shall be spaced not exceeding 150 mm centre to centre and the bars bent to required profile shall be placed symmetrically between two consecutive straight bars. Unless otherwise specified or directed by the Engineer-in-Charge, bars placed alternatively with straight bars shall be bent to form a corrugated profile such that the pitch of the corrugation is 100 to 120 mm and the depth of corrugation is 80 to 100 mm. All the bent bars shall have uniform profile. Straight bar along with the adjoining bent bars on it both sides shall be held in position by passing the bars through holes in the links. Each link shall have three holes and the length of the links shall be such that the distance from the centre of the hole to the nearest edge of the flat is not less than the diameter of the hole. The corner of the links shall be rounded. All links shall be of uniform size and shape. The spacing of the links measured along the straight bar shall be of uniform size and shape. The spacing of the links measured along the straight bar shall be the same as centre to centre distance between two consecutive crests/troughs of the bars bent to the required profile. Each bar and link shall be a continuous single piece without any joint.

ELECTRCAL OPERATION:

Electrically operated for the Rolling Shutters as per statutory norms.

INDUSTRIAL DOORS

Refer document - MAD-appendix 2.104-b-technical data industrial doors - V3

Opening/Closing Speed = 0.12- 0.2 Mtr/Sec

U Value = 1.1 W/m2K as per EN 12667

Air Permeability = Class 2 as per EN 12426

Resistance to Water Penetration = Class 3 as per EN 12425

Wind Class = Class 3 as per EN 12444

Acoustics = 25dB as per EN 717-1

Guide Frame: Specially designed high precision Galvanised steel guide frame with EPDM sealing. Double modular guide frame provides rigidity to door movement and proper fixing to support structure

Door Panel: Thermally efficient, weather proof, finger safe 40 mm thk door panel constructed of Chloroflouro carbide free self extinguishing Polyurethane Foam (PUF) panels sandwiched between 0.5 mm thk GI precoated sheets. Panel height 610 mm, density 40 Kg/ cu mtr. Surface finish Stucco with ribbed profile on outside and plain with stucco inside. Surface colour RAL 9010 from both sides. Optional colour available on request.

Bottom Profile : Precisely designed Galvanised steel bottom profile with provision to fix EPDM sealing for full width of opening

Door Sealings: EPDM soft edge rubber header seal at top panel, Floor seal at bottom edge of panel . Side seals, Intermediate seals between the panels to fill the gap and provide finger trap protection

Hinges with roller assembly: Heavy duty industrial hinges for interlocking two adjacent panels should be provided with lateral rollers with bearings to slide in guide frame.

Roller Shaft assembly : Pre assembled torsion spring shaft assembly with spring break protection. Spring shaft, corner and intermediate bearing and cable drum are pre-assembled as a single unit.

Automation System:

Electric Drive: Electric motor with power ranges from 0.25 Kw to 0.55 Kw and driving torque from 100 Nm to 130 Nm. Operating voltage 400 V , 3 phase, frequency 50 Hz. Duty cycle 60 - 80 %. Protection grade IP 65. Temperature range -20°C to + 60°C.

Temperature range -20°C to + 60°C. Control: Advance control panel, 400 V, 50 / 60 Hz, max current 3.2 A, Duty cycle 60%, Protection class IP 54. Control certified as per DIN EN 13241-1, DIN EN 12453 DIN EN 12445, DIN EN 12978. Temperature ranges from - 10° C to 45 ° C, Air humidity to 80% non condensing.